DNS

Scenario Overview

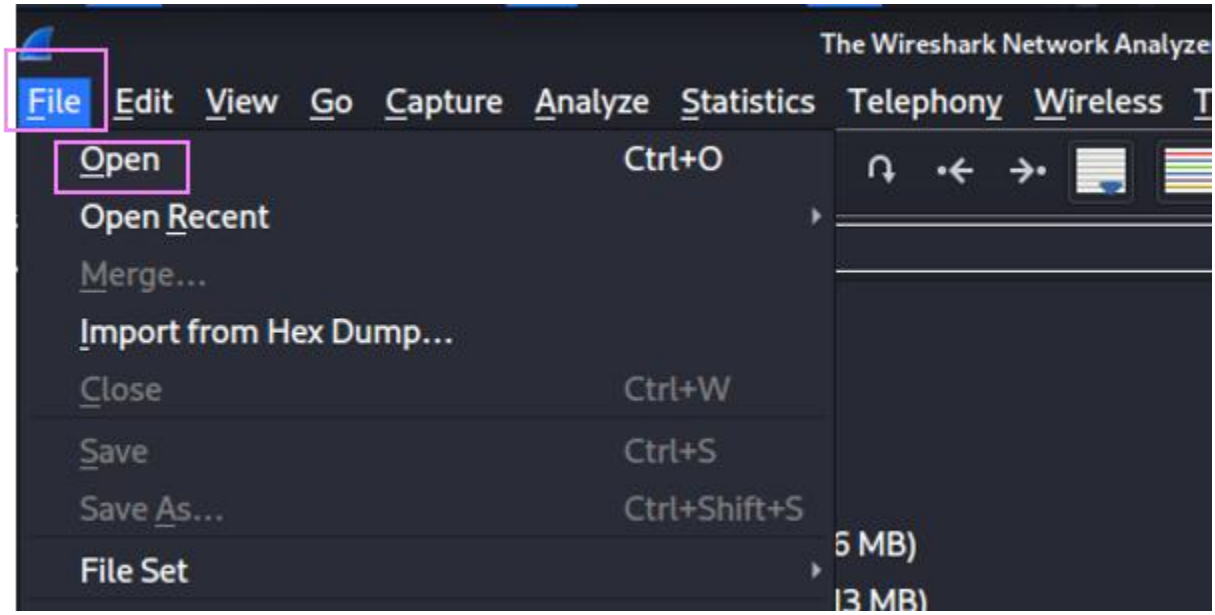
DNS plays a critical role in how users and systems access resources across the internet by translating human-readable domain names into IP addresses. The security team captured a network packet trace containing DNS traffic and suspects that it may reveal unusual or sensitive activity. Your task is to analyze the packet capture, examine the DNS queries and responses, and extract key information such as query type, domain name, record count, TTL values, and associated IP addresses.

Tasks

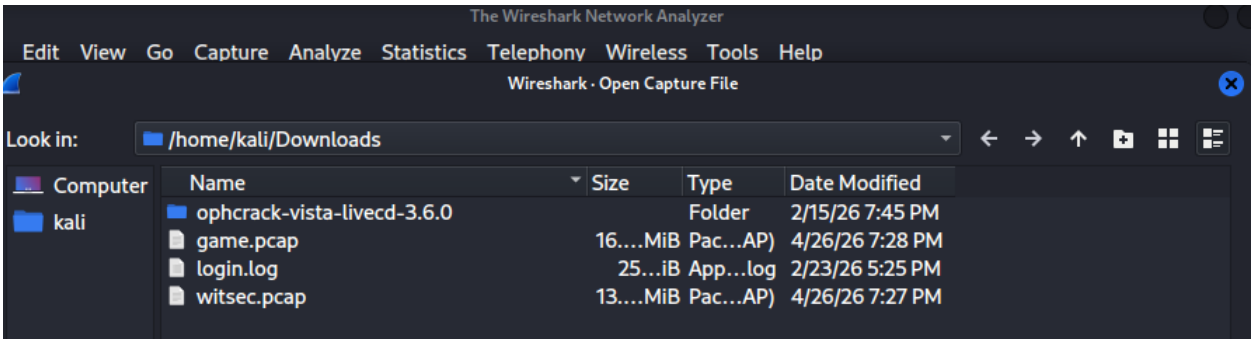
1. **Identify the DNS query type**
    - Open the packet capture and locate the DNS request.
    - Inspect the “Domain Name System” section of the packet details.
    - Determine the query type (e.g., A, AAAA, AXFR, etc.).
2. **Determine the requested domain**
    - Review the DNS query fields.
    - Identify the domain name associated with the request.
3. **Analyze the DNS response**
    - Locate the corresponding DNS response packet.
    - Expand the “Answers” section to determine how many records were returned.
4. **Identify the TTL values**
    - Inspect the DNS answer records.
    - Record the Time-To-Live (TTL) value associated with the returned DNS entries.
5. **Extract the IP address for the specified subdomain**
    - Locate the answer record for the “welcome” subdomain.
    - Identify the IP address returned in the DNS response.

1. What is the type of the DNS query requested?

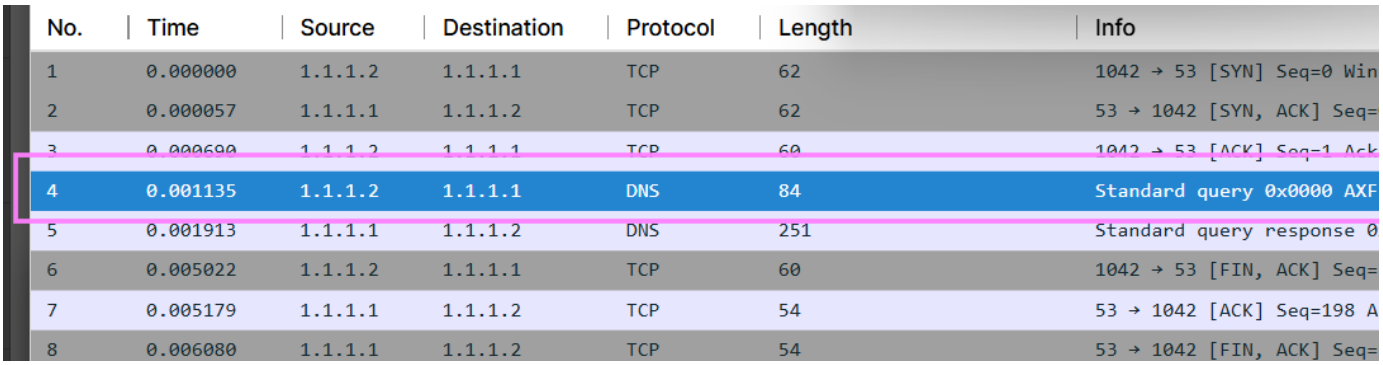
Open Wireshark and load you pcap file by selecting File → Open



Select your pcap file.



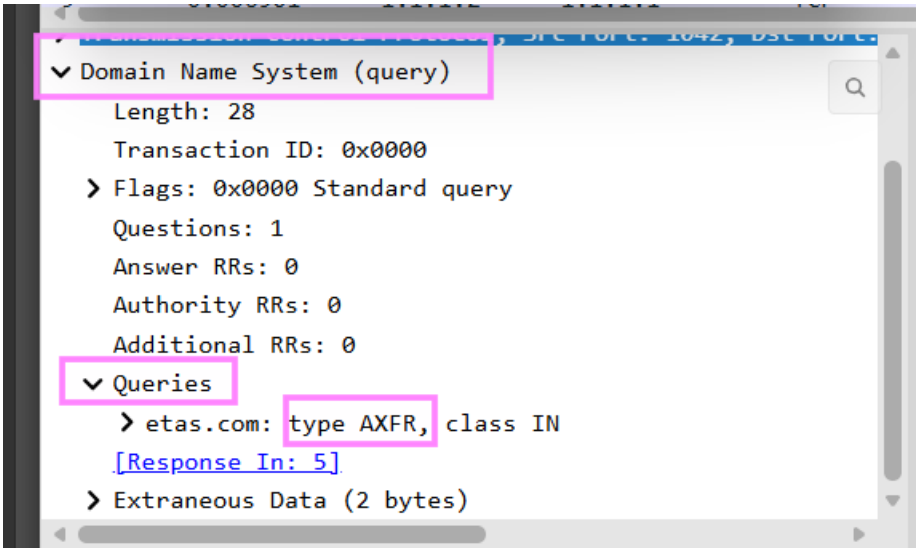
Select the first packet that lists "Standard Query" under the "Info" column (line 4).



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	1.1.1.2	1.1.1.1	TCP	62	1042 → 53 [SYN] Seq=0 Win
2	0.000057	1.1.1.1	1.1.1.2	TCP	62	53 → 1042 [SYN, ACK] Seq=
3	0.000690	1.1.1.2	1.1.1.1	TCP	60	1042 → 53 [ACK] Seq=1 Ack
4	0.001135	1.1.1.2	1.1.1.1	DNS	84	Standard query 0x0000 AXF
5	0.001913	1.1.1.1	1.1.1.2	DNS	251	Standard query response 0
6	0.005022	1.1.1.2	1.1.1.1	TCP	60	1042 → 53 [FIN, ACK] Seq=
7	0.005179	1.1.1.1	1.1.1.2	TCP	54	53 → 1042 [ACK] Seq=198 A
8	0.006080	1.1.1.1	1.1.1.2	TCP	54	53 → 1042 [FIN, ACK] Seq=

Expand the packet dissectors and select the drop down for “Domain Name System.” Select the drop down for “Queries.” Here we see the type of DNS.

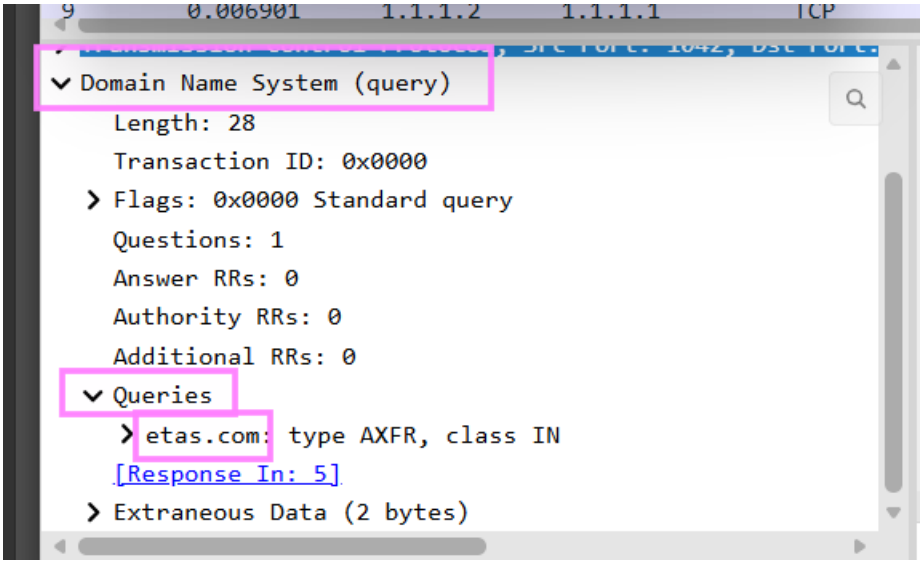
Answer: AXFR



2. What domain was requested?

In the same section, you will see the requested domain next to “type AXFR.”

Answer: etas.com



3. How many items were in the response?

Select the next packet that lists “Standard Query” under the “Info” column (line 5).

4	0.001135	1.1.1.2	1.1.1.1	DNS	84	Standard query 0x0000 AXFR
5	0.001913	1.1.1.1	1.1.1.2	DNS	251	Standard query response 0x
6	0.005022	1.1.1.2	1.1.1.1	TCP	60	1042 → 53 [FIN, ACK] Seq=3

Expand the packet dissectors for “Domain Name System (response)” and select the drop down for “Answers.” You will then see “[Request In: 4].

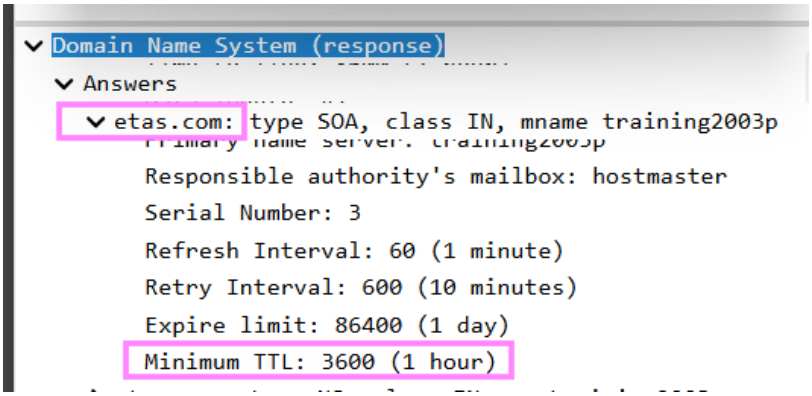
Answer: 4

```
Domain Name System (response)
    Length: 195
    Transaction ID: 0x0000
  > Flags: 0x8080 Standard query response, No error
    Questions: 1
    Answer RRs: 4
    Authority RRs: 0
    Additional RRs: 0
  > Queries
  v Answers
    > etas.com: type SOA, class IN, mname training2003p
    > etas.com: type NS, class IN, ns training2003p
    > welcome.etas.com: type A, class IN, addr 1.1.1.1
    > etas.com: type SOA, class IN, mname training2003p
    [Request In: 4]
```

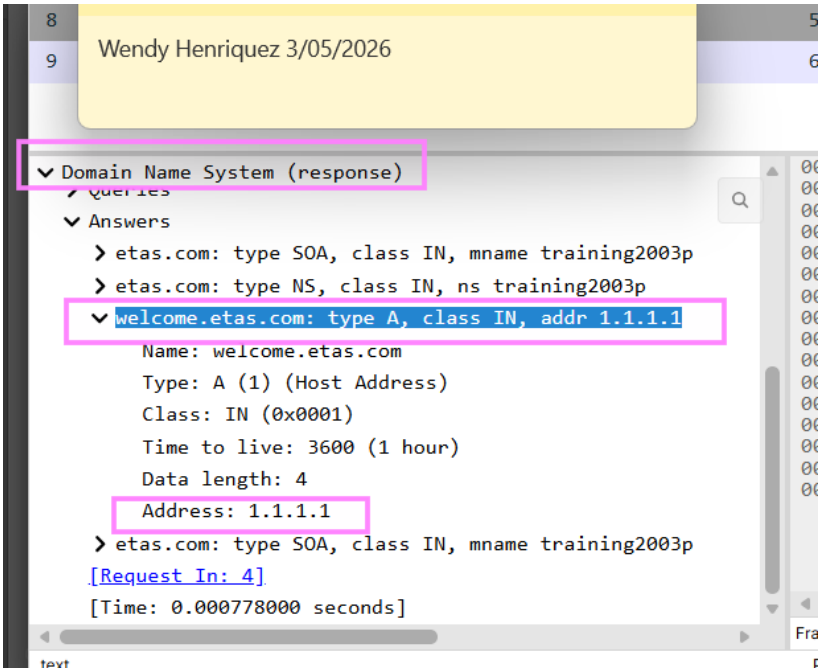
4. What is the TTL for all of the DNS records?

In the same section select the first drop down for “etas.com” and navigate to “Minimum TTL:”

Answer: 3600



5. What is the IP address for the "welcome" subdomain?

In the same section click the drop down for “Welcome.etas.com” and navigate to “Adress.”

Answer: 1.1.1.1